

I/V Stage (OPA1612): Working Document

Digital Audio Player • Rev 10

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1. Summary

Field	Value
Manufacturer / MPN	Texas Instruments, OPA1612AIDRGR (AL-PN #1-104)
Package	8-SON, 3 × 3 mm, dual op-amp
Function	Four current-to-voltage (I/V) converters, one per ES9038Q2M output (DACL, DACLB, DACR, DACRB). Two packages.
Signal chain	I/V (×2 per channel) → differential summing and MFB low-pass → OPA1622 output stage → 3.5 mm jack

Each I/V stage holds its DAC pin at a virtual ground and converts the DAC's output current into a voltage across the feedback resistor R_f . A capacitor C_f across R_f adds a first-order low-pass that starts removing the DAC's ultrasonic noise. The OPA1612 was chosen for its 1.1 nV/√Hz noise, 40 MHz gain-bandwidth and guaranteed operation down to ±2.25 V.

2. Design Requirements

Requirement	Value
DAC output mode	Current mode. ESS specifies its headline figures (128 dB DNR, -120 dB THD+N) in differential current mode.
Supply	±3.3 V analog rails
Maximum headphone output	0.6 V RMS (0.85 V peak), set later in the chain
Gain structure	Run the I/V stage as hot as the rails allow; attenuate in the summing stage
Distortion	Below the DAC's, so the DAC remains the limit
Passband loss	≤ 0.05 dB at 20 kHz from this stage (chain budget 0.1 dB)

Requirement	Value
Constraints	Low noise · Small footprint

3. Pin Configuration

Each OPA1612 package serves one channel: A takes the positive DAC output and B the negative one.

Pin	Name	Connection
1	OUT A	I/V output (IV_L or IV_R); $R_f \parallel C_f$ to -IN A
2	-IN A	DAC output DACL or DACR
3	+IN A	GND ($V_G = 0$ V)
4	V-	-3V3, 0.1 μ F X7R to GND
5	+IN B	GND
6	-IN B	DAC output DACLB or DACRB
7	OUT B	I/V output (IV_LB or IV_RB); $R_f \parallel C_f$ to -IN B
8	V+	+3V3_ANALOG, 0.1 μ F X7R to GND

4. External Components

Ref	Value	Part / Notes	Qty	AL-PN
U	Dual op-amp	TI OPA1612AIDRGR, 8-SON	2	#1-104
R_f	499 Ω , 0.1%, 1/16 W, 0402	YAGEO RT0402BRD07499RL, thin film	4	#1-105
C_f	1.5 nF, 50 V, COG, 1%, 0603	Murata GCM1885C1H152FA16D	4	#1-106
C_{dec}	0.1 μ F X7R, 0402	One per supply pin	4	

0.1% resistors and 1% COG capacitors keep the two legs of each channel matched, so their common DC offset and common-mode noise cancel in the summing stage.

5. Equations and Results

I/V transfer (+IN at $V_G = 0$ V)

$$V_{OUT} = V_G - I_{DAC} R_f$$

DAC full-scale current and current-mode offset

$$I_{FS,pp} = \frac{V_{OPP}}{R_{DAC}} = \frac{0.906 \cdot A_{VCC}}{R_{DAC}} = \frac{2.99 \text{ V}}{774 \Omega} = 3.863 \text{ mA}_{pp}$$

$$I_{DC} = \frac{V_{OCM} - V_G}{R_{DAC}} = \frac{1.65 \text{ V} - 0}{774 \Omega} = 2.13 \text{ mA}$$

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 52 (Analog Output: $R_{DAC} = 774 \Omega \pm 11\%$, $V_{OPP} = 0.906 \times AVCC$, $V_{OCM} = AVCC/2$).

Swing limit (worst case $R_{DAC} = 689 \Omega$)

$$(I_{DC} + I_{AC,pk}) R_f \leq V_{swing} \Rightarrow R_{f,max} = \frac{2.6 \text{ V}}{2.39 \text{ mA} + 2.17 \text{ mA}} = 570 \Omega$$

Source: [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#), Output Voltage vs Output Current: ≈ 0.7 V from each rail at 4.6 mA (measured at ± 15 V).

Feedback-capacitor corner and 20 kHz loss

$$f_c = \frac{1}{2\pi R_f C_f} = \frac{1}{2\pi \cdot 499 \Omega \cdot 1.5 \text{ nF}} = 213 \text{ kHz}$$

$$\text{Loss}(20 \text{ kHz}) = 10 \log_{10} \left(1 + \left(\frac{20 \text{ kHz}}{f_c} \right)^2 \right) = 0.038 \text{ dB}$$

Source: first-order RC low-pass response.

Quantity	Nominal ($R_{\text{DAC}} = 774 \Omega$)	Worst case ($R_{\text{DAC}} = 689 \Omega$)
DC current per pin	2.13 mA	2.39 mA
Audio peak current per pin	1.93 mA	2.17 mA
Maximum R_f (± 2.6 V swing)	640 Ω	570 Ω
I/V output peak per leg ($R_f = 499 \Omega$)	0.96 V	1.08 V
Total I/V output peak (DC + audio)	2.03 V	2.28 V

$R_f = 499 \Omega$ leaves about 0.3 V of worst-case margin to the swing limit. Raising R_f to 570 Ω would improve this stage's signal-to-noise by only $10 \log_{10}(570/499) \approx 0.6$ dB, because signal grows with R_f while R_f 's thermal noise grows with $\sqrt{R_f}$. The DAC's own noise scales with its signal and is unaffected. The DC current is common to both legs of a channel and cancels in the summing stage.

6. Important Values

Item	Value
OPA1612 input voltage noise	1.1 nV/ $\sqrt{\text{Hz}}$ at 1 kHz
OPA1612 gain-bandwidth	40 MHz
OPA1612 distortion	0.000015% (≈ -136 dB) at 1 kHz
OPA1612 supply range	± 2.25 V to ± 18 V (electrical characteristics specified over this range)
OPA1612 quiescent current	3.6 mA per channel
Output swing on ± 3.3 V	$\approx \pm 2.6$ V at 4.6 mA
I/V corner	213 kHz; 0.038 dB loss at 20 kHz

Source: [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#).

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0	2026-09-24	Initial issue: signal chain, DAC output data, candidate evaluation.
1	2026-09-24	OPA1612 selected; OPA1692, OPA4134 and SSM2135 evaluated.

Rev	Date	Change
2 to 5	2026-09-24	Candidate table, LTspice-model requirement and alternative-part links removed; package set to 8-SON.
6	2026-09-24	$V_G = 0\text{ V}$ and R_f selected; operating point added.
7	2026-09-24	$C_f = 1.5\text{ nF}$ COG selected.
8	2026-09-24	$R_f = 499\text{ }\Omega$; swing limit $\pm 2.6\text{ V}$; R_f rationale added.
9	2026-09-24	Component part numbers added.
10	2026-09-25	Reformatted; op-amp in title; requirements and constraints merged; equations typeset; linearity open problem removed.

9. Sources

1. [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#)
2. [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#)